การวิเคราะห์ข้อมูลเชิงสำรวจ
Exploration Data Analysis (EDA)

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Main Steps of Analytics

1. Business Understanding
2. Data Acquisition & Storage
3. **Data Exploration Analysis (EDA) and Data-Preparation**
4. Modeling & Evaluation
   - **Data Mining**: Looking for patterns in data
   - **Data Science**
     - **Machine Learning**:
       Using data to build models and make predictions
     - **Deep Learning, LLM, …**
5. Deployment (e.g., API, dashboard)
6. Monitoring and Feedback

Exploration Data Analysis: EDA

- **Exploratory Data Analysis (EDA)** is the process of exploring and summarizing key characteristics of data using **statistics, data visualization, ... => Understanding Data**
- **Core Components:**
  - Statistical summaries and descriptive analysis
  - Visual representation through graphs and charts
  - Data quality assessment
  - Cleaning insights for next data pre-processing step
  - Advanced techniques ...
- **Key Benefits:**
  - Helps **understand data structure and properties**
  - Provides essential foundation **before building models**
  - Reveals **patterns, outliers, and relationships within the data**
  - Support **decision-making for selecting appropriate analysis methods**

Steps of Exploration Data Analysis **(EDA)**

Phase 1: Data Overview

- Load data and check dimensions (df.shape, df.shape(), df.info())
- Review data types (df.dtypes)
- Generate basic statistical summary such as descriptive stats (df.describe())

Phase 2: Data Quality Assessment

- **Missing Values**: Identify and handle (df.isnull().sum())
- **Outliers**: Detect extreme values (e.g., boxplots, z-scores, IQR)
- **Duplicate Records**: Identify and remove duplicates
- **Irrelevant Features**: Drop un-informative columns (e.g., ID)

Phase 3: Uni-variate Analysis

- **Numerical Features**: Distribution charts such as Histogram, boxplot, …
- **Categorical Features**: Value counts such as bar-plots
- **Temporal Features**: Time-based patterns or seasonality

Phase 4: Bi-variate Analysis & Multi-variate Analysis

- **Interaction between features** (e.g., scatterplots, heatmaps)
- **Relationship with target variable** (e.g., correlation, group means)
- **Advanced analysis**: Segmentation (clustering), Feature Engineering, **OLAP**

Phase 1: Data Overview

Data Overview

- Load data and check dimensions (df.shape, df.shape(), df.info())
- Review data types (df.dtypes)
- Generate basic statistical summary such as descriptive stats (df.describe())

Boston Housing Data with Attribute Description

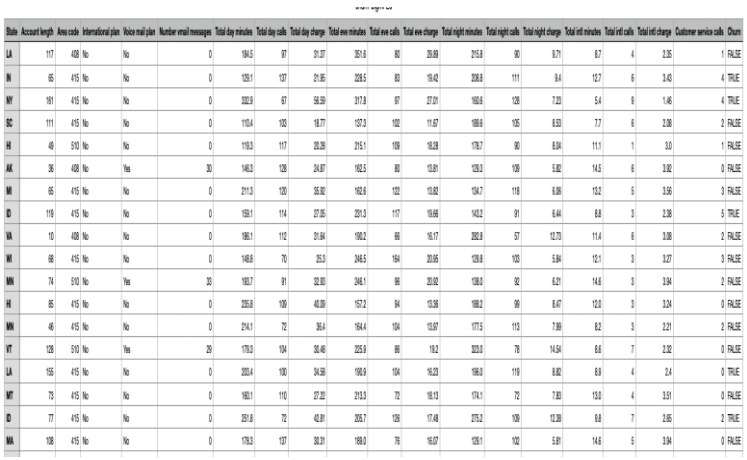
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT	MEDV	CAT. MEDV
0.006	18	2.31	0	0.54	6.58	65.2	4.09	1	296	15.3	397	5	24	0
0.027	0	7.07	0	0.47	6.42	78.9	4.97	2	242	17.8	397	9	21.6	0
0.027	0	7.07	0	0.47	7.19	61.1	4.97	2	242	17.8	393	4	34.7	1
0.032	0	2.18	0	0.46	7.00	45.8	6.06	3	222	18.7	395	3	33.4	1
0.069	0	2.18	0	0.46	7.15	54.2	6.06	3	222	18.7	397	5	36.2	1
0.030	0	2.18	0	0.46	6.43	58.7	6.06	3	222	18.7	394	5	28.7	0
0.088	12.5	7.87	0	0.52	6.01	66.6	5.56	5	311	15.2	396	12	22.9	0
0.145	12.5	7.87	0	0.52	6.17	96.1	5.95	5	311	15.2	397	19	27.1	0
0.211	12.5	7.87	0	0.52	5.63	100	6.08	5	311	15.2	387	30	16.5	0
0.170	12.5	7.87	0	0.52	6.00	85.9	6.59	5	311	15.2	387	17	18.9	0

CRIM per capita crime rate by town
ZN proportion of residential land zoned for lots over 25,000 sq.ft.
INDUS proportion of non-retail business acres per town.
CHAS Charles River dummy variable (1 if tract bounds river; 0 otherwise)
NOX nitric oxides concentration (parts per 10 million)
RM average number of rooms per dwelling
AGE proportion of owner-occupied units built prior to 1940
DIS weighted distances to five Boston employment centres
RAD index of accessibility to radial highways
TAX full-value property-tax rate per $10,000
PTRATIO pupil-teacher ratio by town
B 1000(Bk - 0.63)^2 where Bk is the proportion of blacks by town
LSTAT % lower status of the population
MEDV Median value of owner-occupied homes in $1000

Summary Statistics – Boston Housing

	Average	Median	Min	Max	Std	Count	Countblank
CRIM	3.61	0.26	0.01	88.98	8.60	506	0
ZN	11.36	0.00	0.00	100.00	23.32	506	0
INDUS	11.14	9.69	0.46	27.74	6.86	506	0
CHAS	0.07	0.00	0.00	1.00	0.25	506	0
NOX	0.55	0.54	0.39	0.87	0.12	506	0
RM	6.28	6.21	3.56	8.78	0.70	506	0
AGE	68.57	77.50	2.90	100.00	28.15	506	0
DIS	3.80	3.21	1.13	12.13	2.11	506	0
RAD	9.55	5.00	1.00	24.00	8.71	506	0
TAX	408.24	330.00	187.00	711.00	168.54	506	0
PTRATIO	18.46	19.05	12.60	22.00	2.16	506	0
B	356.67	391.44	0.32	396.90	91.29	506	0
LSTAT	12.65	11.36	1.73	37.97	7.14	506	0
MEDV	22.53	21.20	5.00	50.00	9.20	506	0

Churn Dataset Example



State	Account length	Area code	International plan	Voice mail plan	Number vmail messages	Total day minutes	Total day calls	Total day charge	Total eve minutes	Total eve calls	Total eve charge	Total night minutes	Total night calls	Total night charge	Total intl minutes	Total intl calls	Total intl charge	Customer service calls	Churn
LA	117	408	No	No	0	184.5	97	31.37	351.6	80	29.89	215.8	90	9.71	8.7	4	2.35	1	FALSE
IN	65	415	No	No	0	129.1	137	21.95	228.5	83	19.42	208.8	111	9.4	12.7	6	3.43	4	TRUE
NY	161	415	No	No	0	332.9	67	56.59	317.8	97	27.01	160.6	128	7.23	5.4	9	1.46	4	TRUE
SC	111	415	No	No	0	110.4	103	18.77	137.3	102	11.67	189.6	105	8.53	7.7	6	2.08	2	FALSE
HI	49	510	No	No	0	119.3	117	20.28	215.1	109	18.28	178.7	90	8.04	11.1	1	3.0	1	FALSE
AK	36	408	No	Yes	30	140.3	126	24.87	162.5	80	13.81	129.3	109	5.82	14.5	6	3.92	0	FALSE
MI	65	415	No	No	0	211.3	120	35.92	162.6	122	13.82	134.7	116	6.06	13.2	5	3.56	3	FALSE
ID	119	415	No	No	0	159.1	114	27.05	231.3	117	19.66	143.2	91	6.44	8.8	3	2.38	5	TRUE
VA	10	408	No	No	0	186.1	112	31.64	190.2	66	16.17	282.8	57	12.73	11.4	6	3.08	2	FALSE
WI	68	415	No	No	0	148.8	70	25.3	246.5	164	20.95	139.8	103	6.29	12.1	3	3.27	3	FALSE
MN	74	510	No	Yes	33	160.7	91	32.93	246.1	99	20.92	158.0	92	6.21	14.8	3	3.94	2	FALSE
HI	85	415	No	No	0	235.8	109	40.09	157.2	94	13.36	188.2	99	8.47	12.0	3	3.24	0	FALSE
MN	46	415	No	No	0	214.1	72	36.4	164.4	104	13.97	177.5	113	7.99	8.2	3	2.21	2	FALSE
VT	128	510	No	Yes	29	179.3	104	30.48	225.9	86	19.2	323.0	78	14.54	8.6	7	2.32	0	FALSE
LA	155	415	No	No	0	203.4	100	34.58	166.9	104	14.23	196.0	119	8.82	8.9	4	2.4	0	TRUE
MT	73	415	No	No	0	160.1	110	27.22	213.3	72	18.13	174.1	72	7.83	13.0	4	3.51	0	FALSE
ID	77	415	No	No	0	251.8	72	42.81	205.7	135	17.48	273.2	109	12.29	9.8	7	2.65	2	TRUE
MA	108	415	No	No	0	178.5	137	30.31	198.0	78	16.07	129.1	102	5.81	14.6	5	3.94	0	FALSE

- ***State***: Categorical, for the 50 states and the District of Columbia
- ***Account Length***: Integer-valued, how long account has been active
- ***Area code***: Categorical
- ***Phone Number***: Essentially a surrogate for customer ID
- ***International Plan***: Dichotomous categorical, yes or no
- ***Voice Mail Plan***: Dichotomous categorical, yes or no
- ***Number of Voice Mail Messages***: Integer-valued
- ***Total Day Minutes***: Continuous, minutes customer used service during the day
- ***Total Day Calls***: Integer-valued
- ***Total Day Charge***: Continuous, perhaps based on above two variables
- ***Total Eve Minutes***: Continuous, minutes customer used service during the evening
- ***Total Eve Calls***: Integer-valued
- ***Total Eve Charge***: Continuous, perhaps based on above two variables
- ***Total Night Minutes***: Continuous, minutes customer used service during the night
- ***Total Night Calls***: Integer-valued
- ***Total Night Charge***: Continuous, perhaps based on above two variables
- ***Total International Minutes***: Continuous, minutes customer used service to make international calls
- ***Total International Calls***: Integer-valued
- ***Total International Charge***: Continuous, perhaps based on above two variables
- ***Number of Calls to Customer Service***: Integer-valued
- *Churn*: Target. Indicator of whether the customer has left the company (True or False)

Summarization & Visualization of the *churn* data set

Field	Sample Graph	Type	Min	Max	Mean	Std. Dev	Skewn...	Median	Mode	Unique	Valid
State		Set	--	--	--	--	--	--	WV	51	3333
Account Length		Range	1	243	101.065	39.822	0.097	101	105	--	3333
Area Code		Set	408	510	--	--	--	--	415	3	3333
Intl Plan		Flag	--	--	--	--	--	--	no	2	3333
VMail Plan		Flag	--	--	--	--	--	--	no	2	3333
VMail Message		Range	0	51	8.099	13.688	1.265	0	0	--	3333
Day Mins		Range	0.000	350.800	179.775	54.467	-0.029	179.400	154.000*	--	3333
Day Calls		Range	0	165	100.436	20.069	-0.112	101	102	--	3333
Day Charge		Range	0.000	59.640	30.562	9.259	-0.029	30.500	26.180*	--	3333

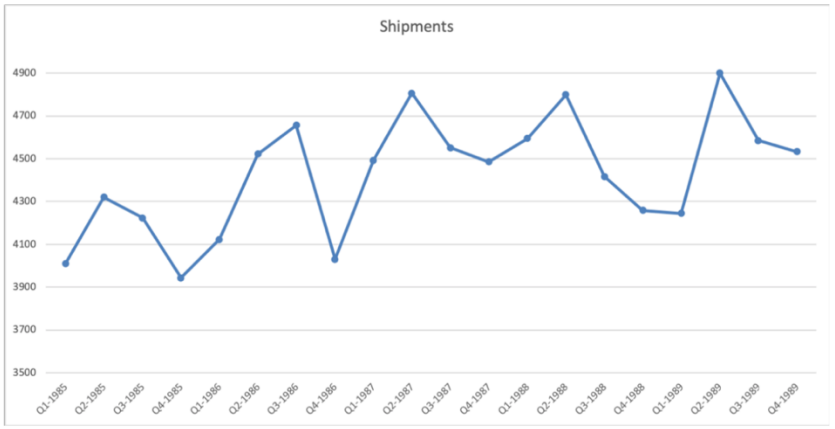
(a)

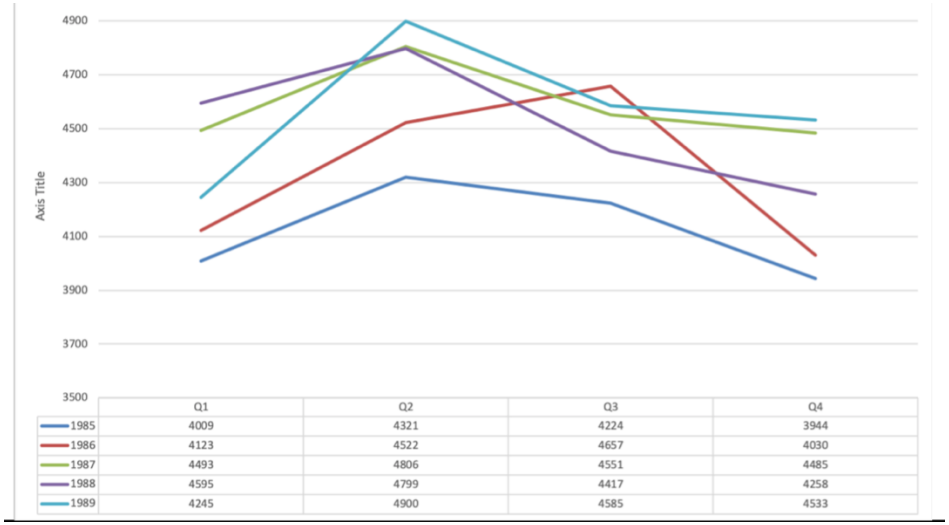
Field	Sample Graph	Type	Min	Max	Mean	Std. Dev	Skewn...	Median	Mode	Unique	Valid
Eve Mins		Range	0.000	363.700	200.980	50.714	-0.024	201.400	169.900	--	3333
Eve Calls		Range	0	170	100.114	19.923	-0.056	100	105	--	3333
Eve Charge		Range	0.000	30.910	17.084	4.311	-0.024	17.120	14.250*	--	3333
Night Mins		Range	23.200	395.000	200.872	50.574	0.009	201.200	188.200*	--	3333
Night Calls		Range	33	175	100.108	19.569	0.032	100	105	--	3333
Night Charge		Range	1.040	17.770	9.039	2.276	0.009	9.050	9.450*	--	3333
Intl Mins		Range	0.000	20.000	10.237	2.792	-0.245	10.300	10.000	--	3333
Intl Calls		Range	0	20	4.479	2.461	1.321	4	3	--	3333
Intl Charge		Range	0.000	5.400	2.765	0.754	-0.245	2.780	2.700	--	3333
CustServ Calls		Range	0	9	1.563	1.315	1.091	1	1	--	3333
Churn		Flag	--	--	--	--	--	--	False	2	3333

(b)

Understanding data via Visualization

Quarter	Shipments
Q1-1985	4009
Q2-1985	4321
Q3-1985	4224
Q4-1985	3944
Q1-1986	4123
Q2-1986	4522
Q3-1986	4657
Q4-1986	4030
Q1-1987	4493
Q2-1987	4806
Q3-1987	4551
Q4-1987	4485
Q1-1988	4595
Q2-1988	4799
Q3-1988	4417
Q4-1988	4258
Q1-1989	4245
Q2-1989	4900
Q3-1989	4585
Q4-1989	4533





Data example: Shipments of Household Appliances

Basic Plots for EDA

Basic Plots

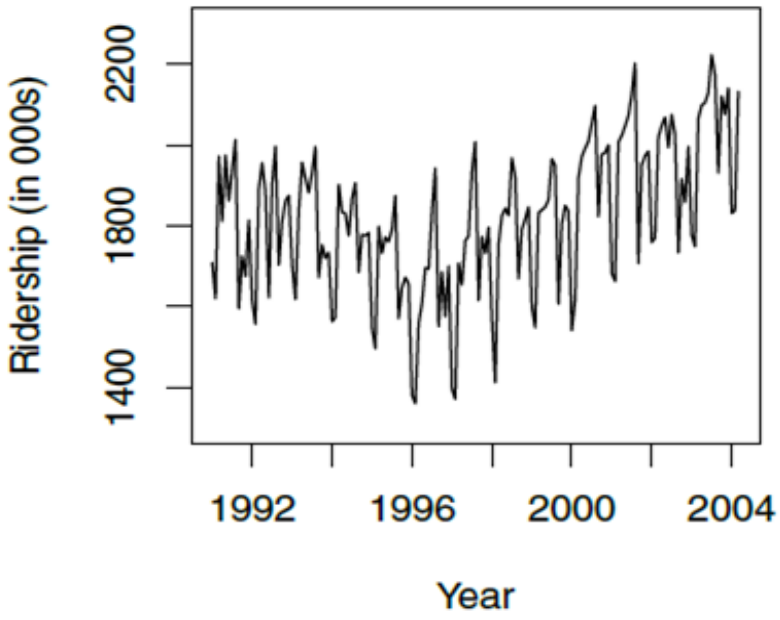
- Line Graphs
- Bar Charts
- Scatterplots
- Etc.

Distribution Plots

- Boxplots
- Histograms
- Etc.

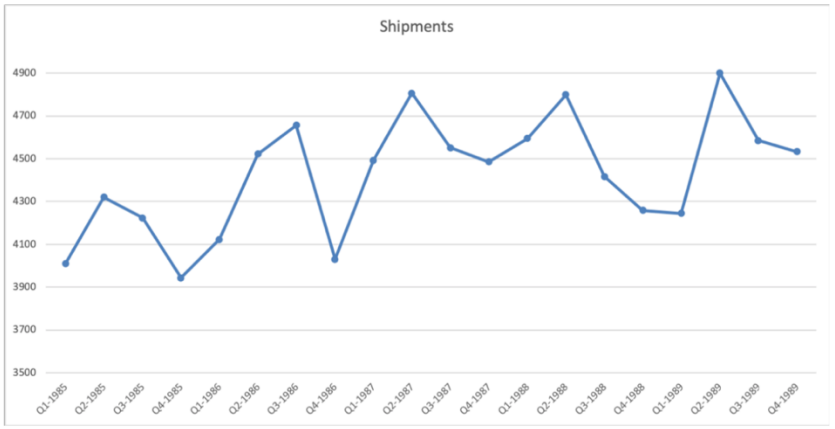
Line Plot

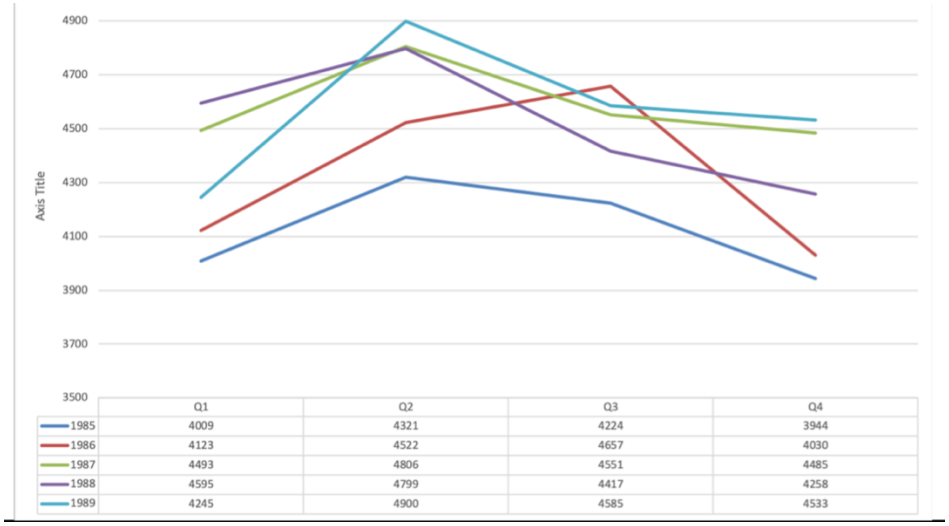
- A line plot, also known as a line chart or a line graph, is a type of data visualization that represents data points as connected line segments.
- It is commonly used to show the relationship or trends between two variables over a continuous period or sequence.
- The horizontal axis (x-axis) typically represents the **independent variable, such as time**, while **the vertical axis (y-axis) represents the dependent variable**, such as a numerical measurement or value.
- Line Plot for Time Series
  - Example displays a line chart for the time series of monthly railway passengers.
  - Line graphs are used primarily for showing time series.



Line Plot

Quarter	Shipments
Q1-1985	4009
Q2-1985	4321
Q3-1985	4224
Q4-1985	3944
Q1-1986	4123
Q2-1986	4522
Q3-1986	4657
Q4-1986	4030
Q1-1987	4493
Q2-1987	4806
Q3-1987	4551
Q4-1987	4485
Q1-1988	4595
Q2-1988	4799
Q3-1988	4417
Q4-1988	4258
Q1-1989	4245
Q2-1989	4900
Q3-1989	4585
Q4-1989	4533

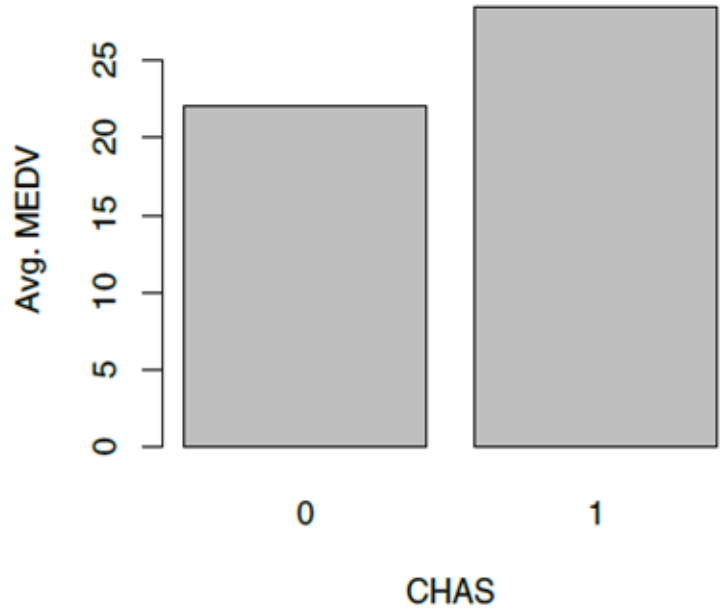




Data example: Shipments of Household Appliances

Bar Chart for Categorical Variable

- Bar charts are useful for comparing a single statistic (e.g., average, count, percentage) across groups.
- The height of the bar (or length, in a horizontal display) represents the value of the statistic (e.g., average, count, percentage).
- Different bars correspond to different groups.

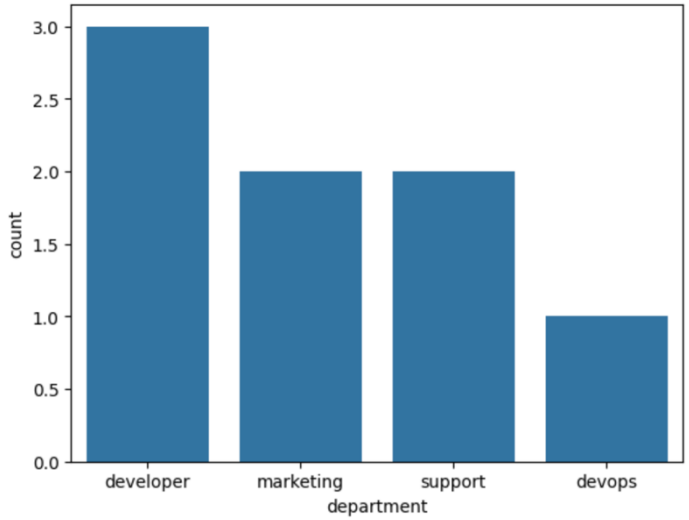


Categorical Frequency: Count frequency of each category

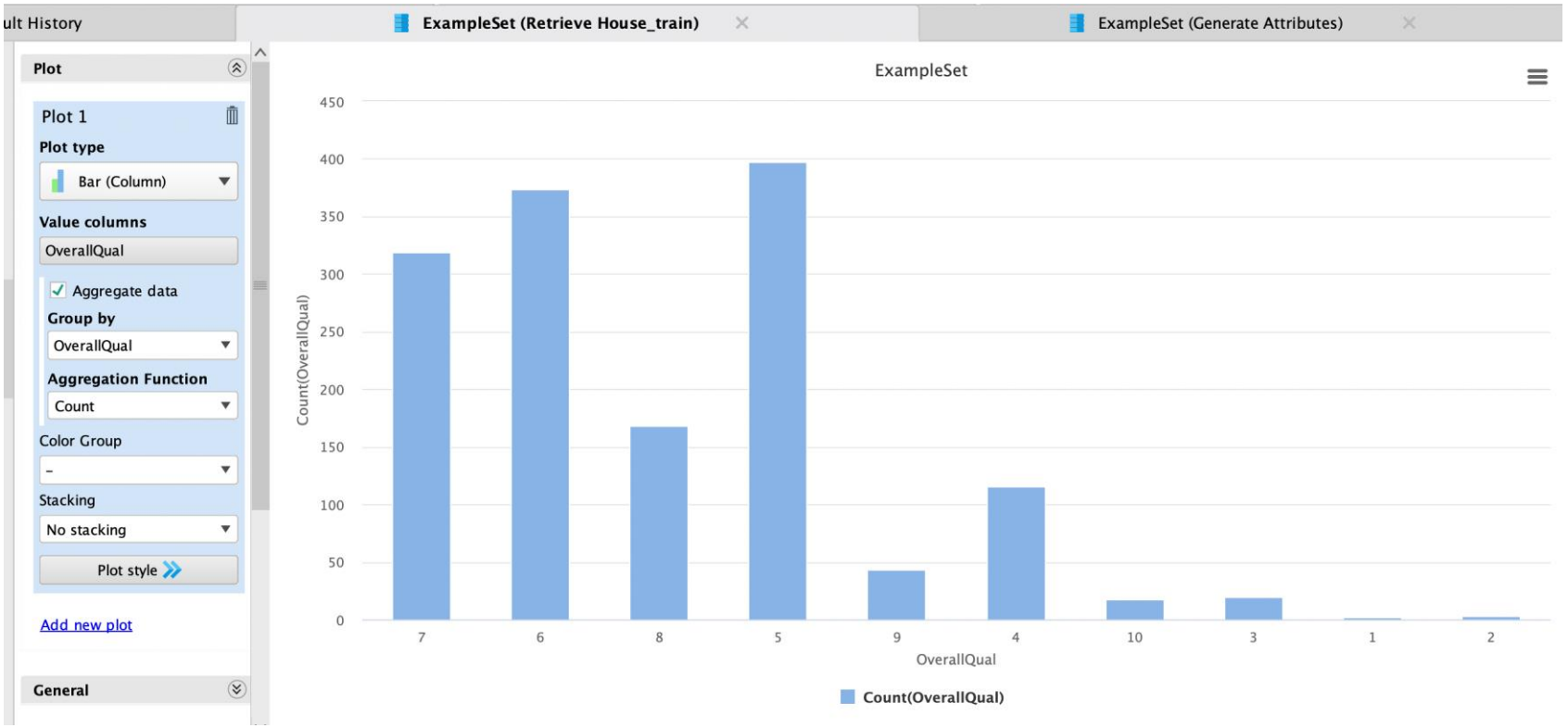
	eid	name	gender	department	age	salary	working_years	birth_place
0	E011	Anda	female	developer	39	64200.0	4	Bangkok
1	E012	Bordin	male	developer	25	48700.0	2	Phuket
2	E013	Chantana	female	developer	29	45500.0	3	Chonburi
3	E014	Donlaya	female	marketing	39	72600.0	8	Chonburi
4	E015	Ekkasit	male	marketing	37	80500.0	7	Suphanburi

```
[5] sns.countplot(data=df, x="department")
```

```
<Axes: xlabel='department', ylabel='count'>
```

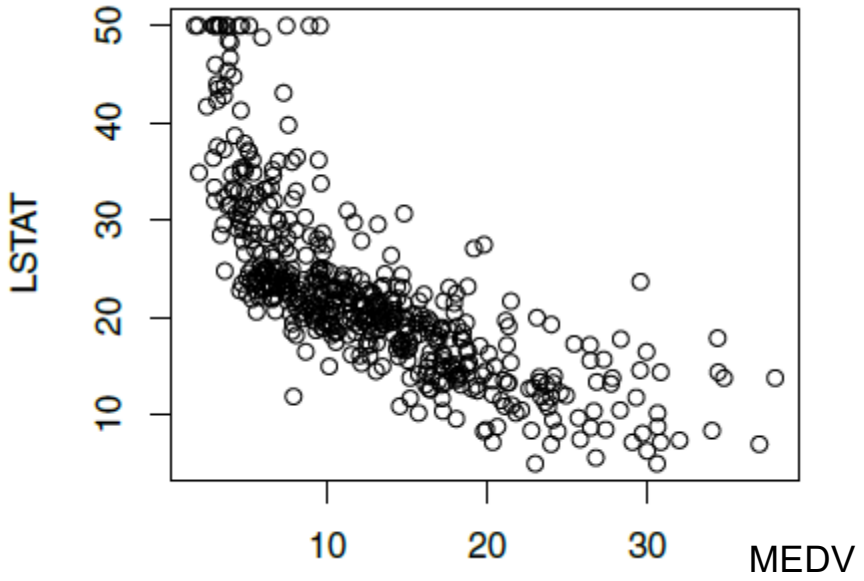


Categorical Frequency: Count frequency of each category



Scatterplot

- Displays relationship between two numerical variables
- Example, it displays a scatterplot of MEDV versus LSTAT.
- This is an important plot in the prediction task.

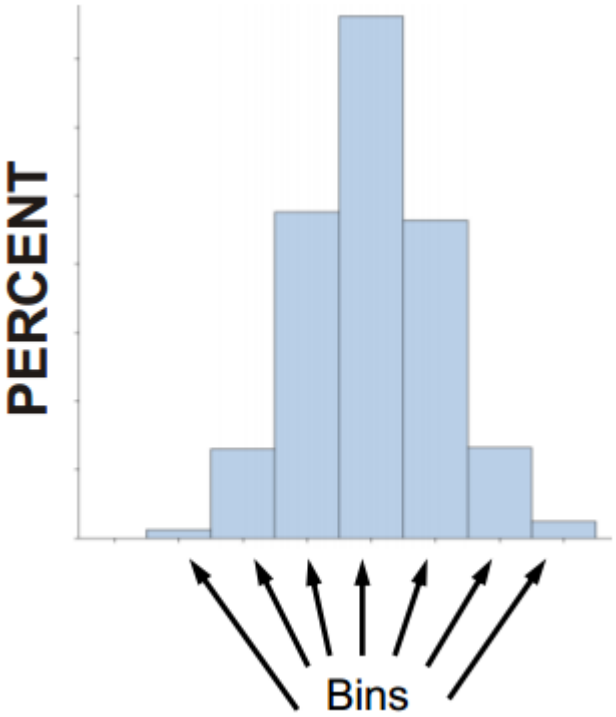


Distribution Plots

- Display “how many” of each value occur in a data set
- Or, for continuous data or data with many possible values, “how many” values are in each of a series of **ranges or “bins”**
- The **boxplot** and the **histogram** are two plots that display the **entire distribution of a numerical variable**.
- Distribution plots are useful for determining potential **data science methods and variable transformations**.
- For example, **skewed numerical variables might warrant transformation** (e.g., moving to a logarithmic scale) if used in methods that assume normality (e.g., linear regression, discriminant analysis).

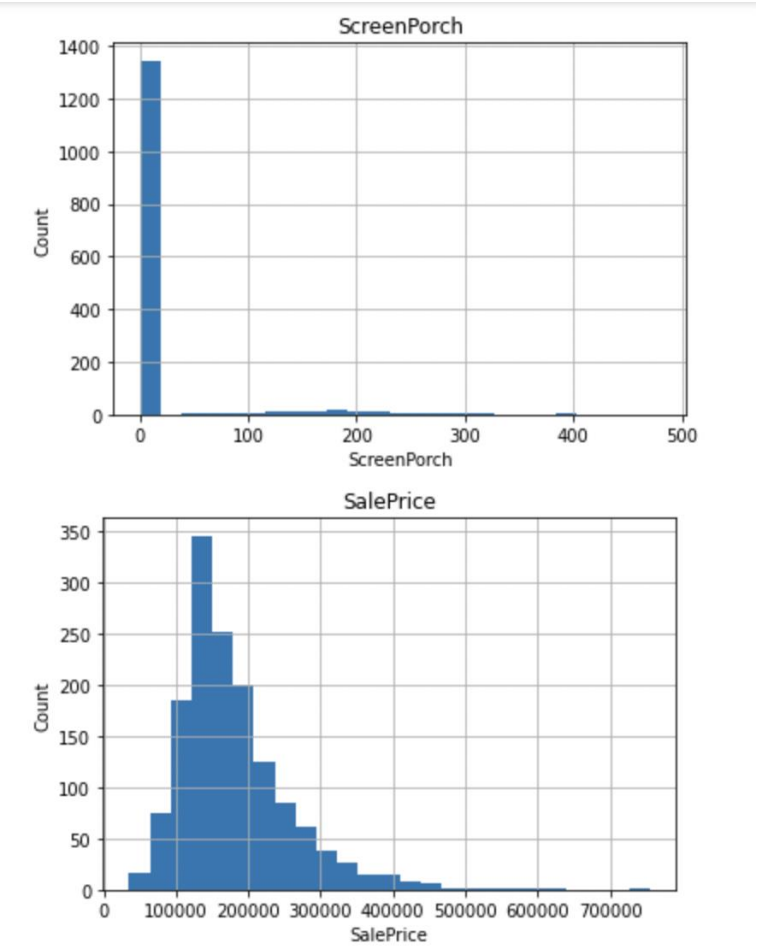
Histograms

- A histogram represents the distribution of a continuous variable.
- It displays the frequency or count of data points falling within specific intervals, known as bins or buckets, along a single axis.
- It divides the range of values into bins and displaying the frequency or count of data points within each bin
- A histogram represents the frequencies of all x values with a series of vertical connected bars.
- Each bar in the histogram represents a group of values (a bin).
- The height of the bar represents the frequency or percent of values in the bin.



Continuous variables: Distribution Plots

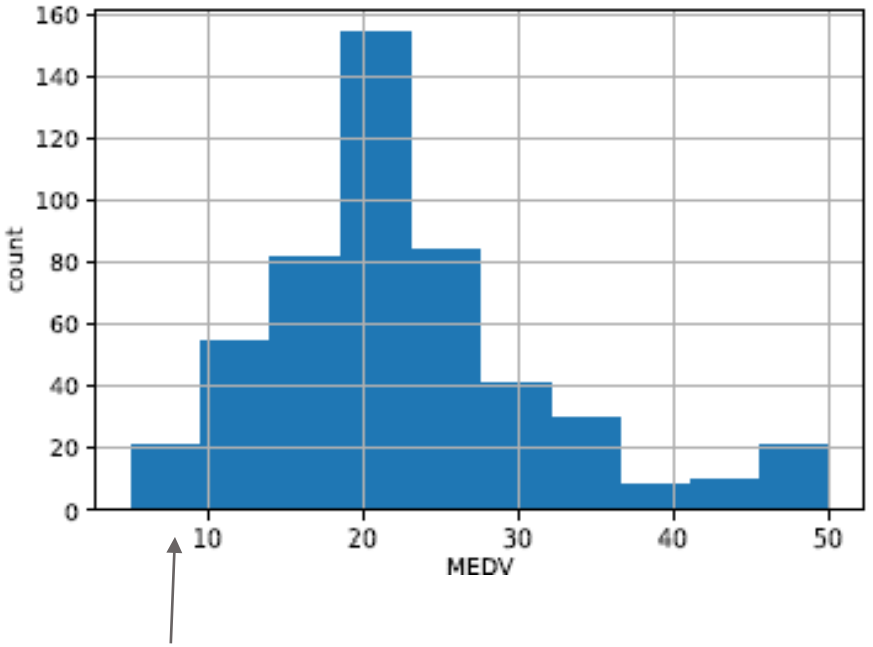
- Display “how many” of each value occur in a data set
- Or, for continuous data or data with many possible values, “how many” values are in each of a series of ranges or “bins”
- For supervised learning:
  - Determine potential data analytics techniques
  - Variable transformation



Histograms

Boston Housing example:

Histogram shows the distribution of the outcome variable (median house value)

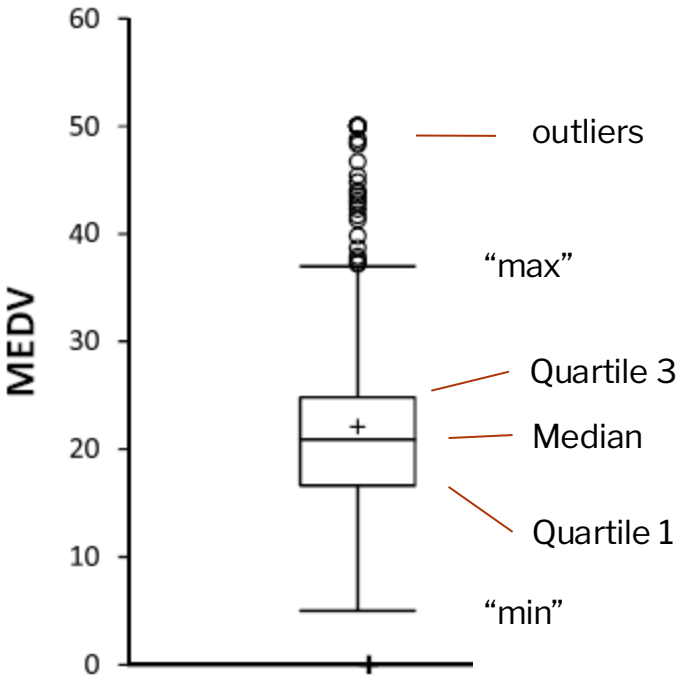


About 20 neighborhoods had a median house value in the lowest bin, about $4000K to $9500 (these data are from mid-20th century)

```
## histogram of MEDV
ax = housing_df.MEDV.hist()
ax.set_xlabel('MEDV'); ax.set_ylabel('count')
```

Boxplot

- A boxplot is a graphical representation of the distribution of a dataset.
- It displays key summary statistics and provides a visual depiction of the variability, skewness, and potential outliers within the data.



Phase 1: Statistical Summary of Data

- **Overview of Data Quality :**
  - Number of records and missing values
  - Detection of noises and outliers
- **Measures of Central Tendency:**
  - **Mean:** Sum of all values ÷ Number of values
  - **Median:** Middle value when data is sorted
  - **Mode:** Most frequently occurring value
- **Measures of Dispersion:**
  - **Range:** Maximum – Minimum value
  - **Variance (σ^2):** Average of squared differences from the mean
  - **Standard Deviation (σ):** Square root of variance
  - **Coefficient of variation**
- **Distribution Analysis:**
  - **Quartiles:** Q1 (25%), Q2 (50% - Median), Q3 (75%)
  - **IQR (Interquartile Range):** Q3 – Q1
  - **Skewness** when close to 0, suggesting a nearly normal distribution
    - |Skewness| < 0.5: เบ้เล็กน้อย (Nearly symmetric)
    - 0.5 ≤ |Skewness| < 1: เบ้ปานกลาง (Moderately skewed)
    - |Skewness| ≥ 1: เบ้มาก (Highly skewed)

Phase 2 : Data Quality Assessment

Phase 2: Data Quality Assessment

- **Missing Values**: Identify and handle (df.isnull().sum())
- **Outliers**: Detect extreme values (e.g., boxplots, z-scores, IQR)
- **Duplicate Records**: Identify and remove duplicates
- **Irrelevant Features**: Drop uninformative columns (e.g., ID)

What is missing value?

Missing values features: are features having missing values

Name	Weight	Gender	Play Cricket/ Not
Mr. Amit	58	M	Y
Mr. Anil	61	M	Y
Miss Swati	58	F	N
Miss Richa	55	?	Y
Mr. Steve	55	M	N
Miss Reena	64	F	Y
Miss Rashmi	57	?	Y
Mr. Kunal	57	M	N

- "I Don't Know"
- Refusals to respond
- Questions left blank
- Failed measurement
- etc.

Missing Values Analysis & Handling

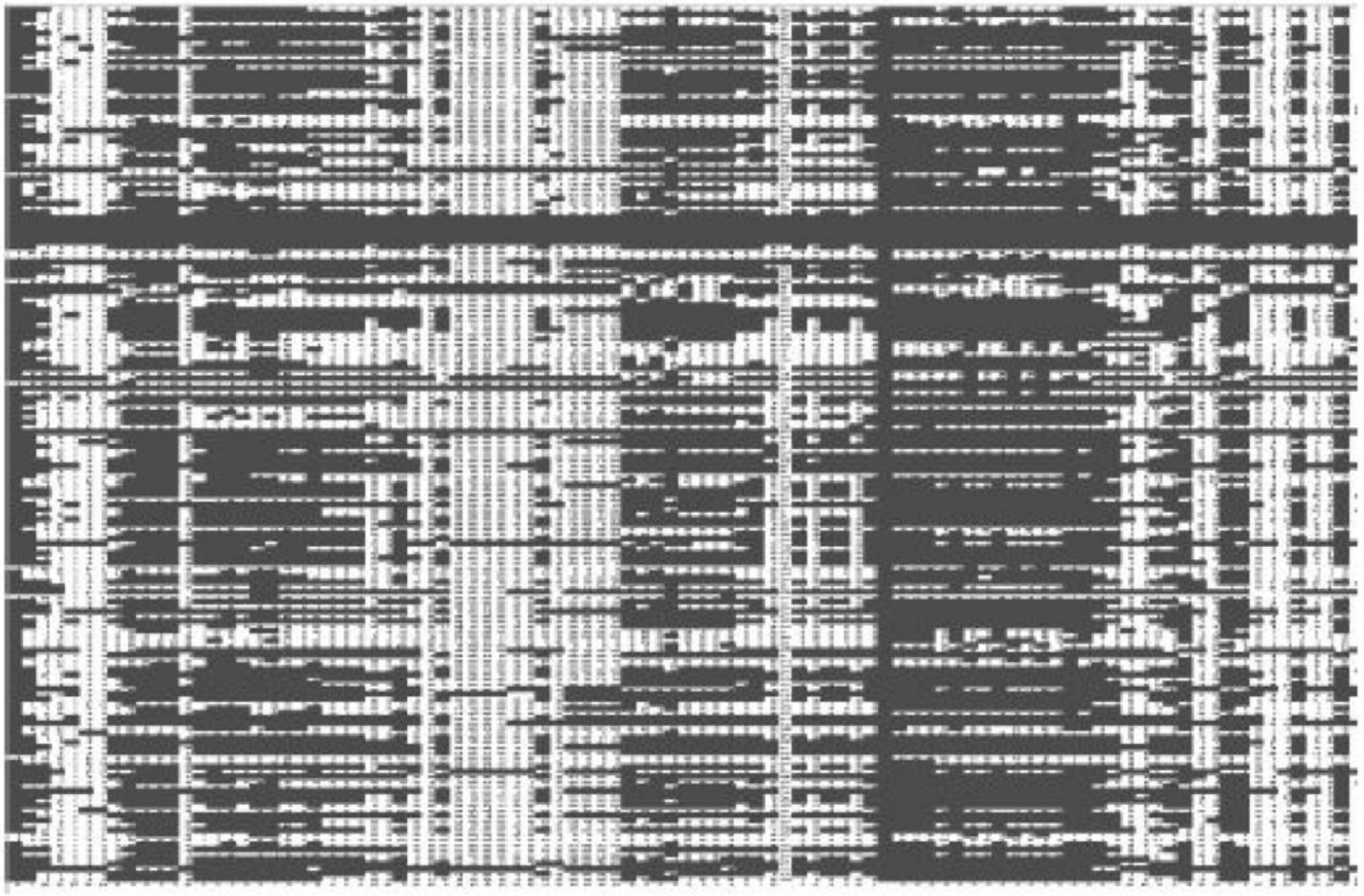
- **Detection & Assessment:**
  - **Check percentage:** df.isnull().sum() / len(df) * 100
  - **Count missing:** df.isnull().sum()
  - **Visualize patterns:** Heatmap sns.heatmap(df.isnull()) or Missing matrix msno.matrix(df)
  - **List affected features** and their missing percentages
- **Decision Framework:**
  - **Rule of Thumb:** Consider dropping columns with >50% missing values
  - **Evaluate impact** on model performance before elimination
- **Handling Strategies:**
  - **Delete:** If missing <5% of total data
  - **Impute:** Mean/Median/Mode, …
  - **Advanced Methods:** KNN Imputation, Multiple Imputation
- **Key Principle:** Always analyze the relationship between missing features and target variable before deciding to eliminate them

Heat Maps

- Heat maps are data visualizations that use color intensity to represent values within a matrix or table.
- They are effective in displaying complex data patterns, relationships, and variations in a visually intuitive way.
- In a data analytics context, heatmaps are especially useful for two purposes: for visualizing **correlation** tables and for visualizing **missing values** in the data.

Heatmap of Missing Values

BLACK DENOTES MISSING VALUE



Missing Values

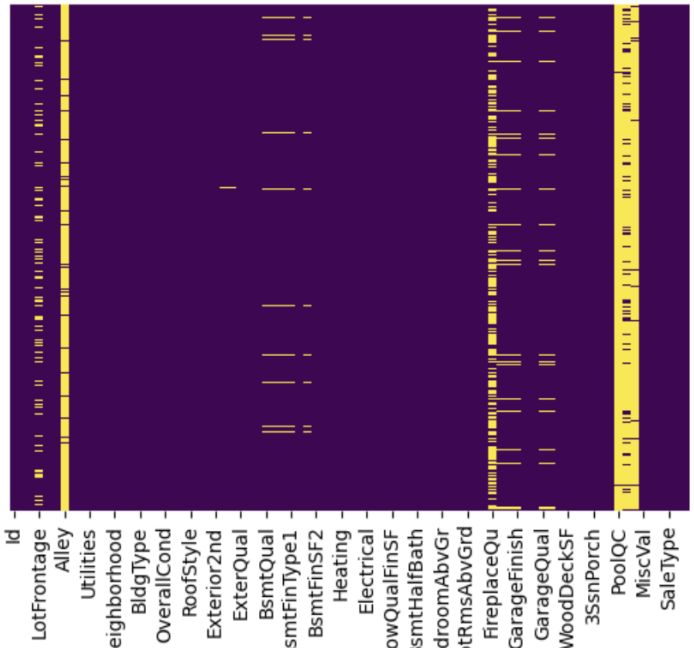
```
[6]  dataset=pd.read_csv('train.csv')

     ## print shape of dataset with nb-rows and nb-columns
     print(dataset.shape)
     print(dataset.dtypes)
```

```
(1460, 81)
Id                int64
MSSubClass        int64
MSZoning         object
LotFrontage     float64
LotArea           int64
                 ...
MoSold            int64
YrSold            int64
SaleType         object
SaleCondition    object
SalePrice         int64
Length: 81, dtype: object
```

```
[4]  sns.heatmap(dataset.isnull(), yticklabels=False, cbar=False, cmap='viridis')
```

```
<Axes: >
```



Outlier Detection

- **What are Outliers?**
  - Data points that differ significantly from the majority of the dataset
- **Detection Methods**
  - **1. Statistical Methods:**
    - **IQR Method**: A data point is an **outlier** if $x < Q1 - 1.5(IQR)$ or $x > Q3 + 1.5(IQR)$
    - **Z-Score**: …
  - **2. Visual Methods:**
    - **Boxplot**: Points outside the whiskers
    - **Scatter Plot**: Points distant from the main cluster
    - **Histogram**: Bars separated from main distribution
  - **Handling Strategies**
    - **Keep**: If valid business case or meaningful extreme values
    - **Remove**: If data entry errors or measurement mistakes
    - **Transform**: Log, Square root, or other transformations

Steps of Exploration Data Analysis **(EDA)**

Phase 1: Data Overview

- Load data and check dimensions (df.shape, df.shape(), df.info())
- Review data types (df.dtypes)
- Generate basic statistical summary such as descriptive stats (df.describe())

Phase 2: Data Quality Assessment

- **Missing Values**: Identify and handle (df.isnull().sum())
- **Outliers**: Detect extreme values (e.g., boxplots, z-scores, IQR)
- **Duplicate Records**: Identify and remove duplicates
- **Irrelevant Features**: Drop un-informative columns (e.g., ID)

Phase 3: Uni-variate Analysis

- **Numerical Features**: Distribution charts such as Histogram, boxplot, …
- **Categorical Features**: Value counts such as bar-plots
- **Temporal Features**: Time-based patterns or seasonality

Phase 4: Bi-variate Analysis & Multi-variate Analysis

- **Interaction between features** (e.g., scatterplots, heatmaps)
- **Relationship with target variable** (e.g., correlation, group means)
- **Advanced analysis**: Segmentation (clustering), Feature Engineering, **OLAP**

Uni-variate, Bi-variate, Multi-variate Analysis

Phase 3: Uni-variate Analysis

- **Numerical Features**: Histogram, boxplot, distribution
- **Categorical Features**: Value counts, bar plots
- **Temporal Features**: Time-based patterns or seasonality

Phase 4: Bi-variate & Multi-variate Analysis

- **Relationship with target variable** (e.g., correlation, group means)
- **Interaction between features** (e.g., scatterplots, heatmaps)
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Uni-variate Analysis

Numerical Variables

- **Distribution Shape:**
  - **Normal**: Bell curve (Mean ≈ Median)
  - **Skewed Right**: Long tail (Mean > Median)
  - **Skewed Left**: Long tail (Mean < Median)
  - **...**
- **Key Metrics:**
  - **Skewness, Kurtosis**:

Categorical Variables

- **Frequency Analysis:**
  - **Value Counts**: count frequency of each category
  - **Relative Frequency**: ration (%)
  - **Cardinality**: count unique values
- **oo** :
  - **Low Cardinality**: < 10 categories (OK)
  - **High Cardinality**: > 50 categories (quite high)

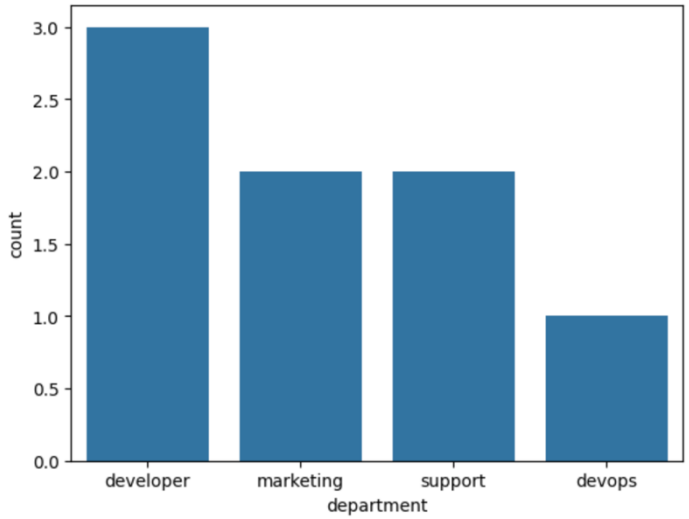
Temporal Variables

Univariate Analysis: Categorical distribution

	eid	name	gender	department	age	salary	working_years	birth_place
0	E011	Anda	female	developer	39	64200.0	4	Bangkok
1	E012	Bordin	male	developer	25	48700.0	2	Phuket
2	E013	Chantana	female	developer	29	45500.0	3	Chonburi
3	E014	Donlaya	female	marketing	39	72600.0	8	Chonburi
4	E015	Ekkasit	male	marketing	37	80500.0	7	Suphanburi

```
[5] sns.countplot(data=df, x="department")
```

```
<Axes: xlabel='department', ylabel='count'>
```



Bivariate Analysis: Numerical vs Numerical Analysis

- **Primary Objectives:**
  - Identify strength and direction of relationships
  - Analyze how changes in one variable correspond to changes in another
- **Key Methods:**
- **Correlation Analysis**
- Quantifies the linear relationship between two numerical variables using correlation coefficients (like Pearson's r).
- **Visualization Techniques:**
- **Scatter Plot**
  - Best for examining individual variable pairs
  - Helps identify patterns and trends
  - Shows the nature of the relationship (linear, non-linear, etc.)
- **Correlation Heatmap**
  - Displays correlation coefficients in a matrix format
  - Color-coded for easy interpretation
  - Excellent for examining multiple variable relationships simultaneously
- **Pairplot**
  - Creates scatter plots for every pair of variables
  - Comprehensive view of all bivariate relationships
  - Often includes histograms on the diagonal

Bivariate Analysis: Numerical vs Numerical Analysis

- **Correlation Analysis**
  - Measures the strength and direction of the linear relationship between two numerical variables, represented by a correlation coefficient (ranging from -1 to 1).
  - A **correlation coefficient** close to 1 or -1 indicates a strong relationship, while a coefficient close to 0 indicates a weak relationship.
- **Visualization**
  - **Scatter Plot**
    - Each point represents an observation in the dataset.
    - The pattern of the points can indicate the nature of the relationship (e.g., linear, non-linear).
  - **Seaborn Pairplot**
    - Visualizes pairwise relationships between multiple numerical variables in a dataset.
    - **Scatter plots for each pair**.
  - **Correlation Heatmap**
    - visualizes the pairwise correlation coefficients between numerical variables using colors. This makes it easy to spot strong positive or negative correlations => Analyze multi-variable relationships together !

Bivariate Analysis: Numerical vs Numerical Analysis

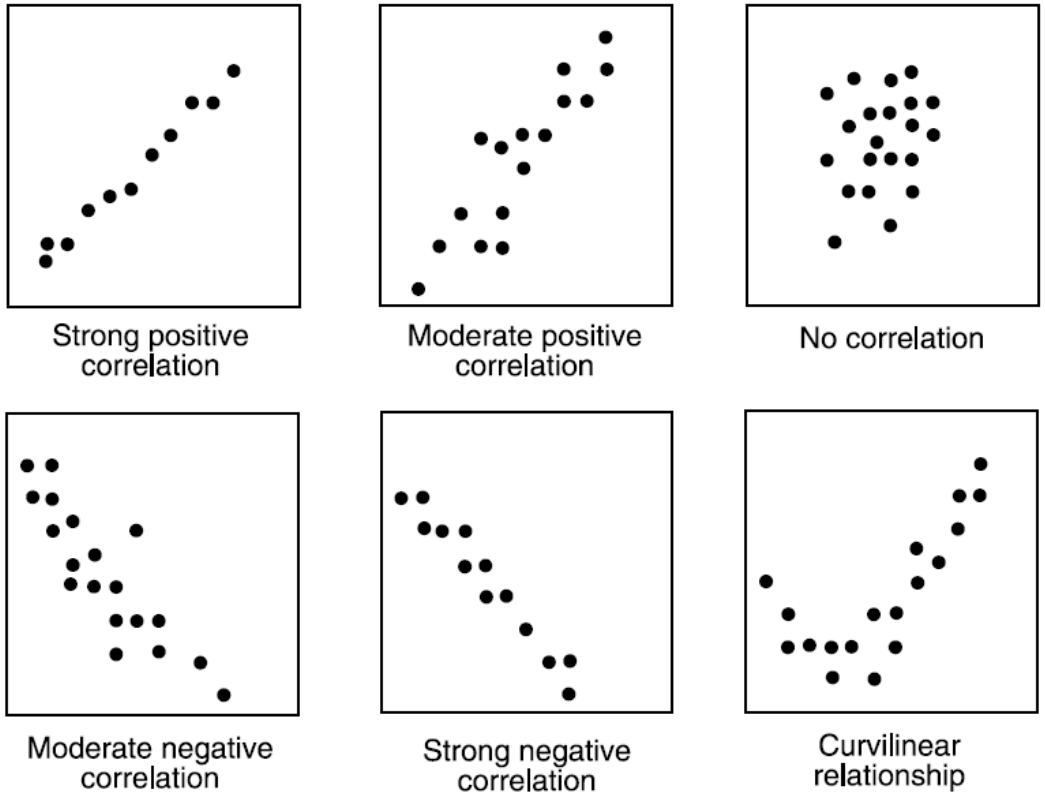
There is Correlation between them

When two variables move together,
we say they are *correlated*.

Bivariate Analysis: Numerical vs Numerical Analysis

Scatter Plot

- Show relationship between two variables
- Not indicates the strength of relationship amongst them.



Relationship between (numerical - numerical) variables

Pearson product moment Correlation coefficient (r):

$$r = \frac{\text{Covariance}(X,Y)}{\text{SQRT}(\text{Variance}(X)*\text{Variance}(Y))}$$

$$\text{Covariance}(X,Y) = \frac{\Sigma(X-\bar{X})(Y-\bar{Y})}{n-1}$$

$$\text{Variance}(X) = \frac{\Sigma(X-\bar{X})^2}{n-1} \qquad \text{Variance}(Y) = \frac{\Sigma(Y-\bar{Y})^2}{n-1}$$

Relationship between (numerical - numerical) variables

Pearson Correlation Coefficient (r):

- r = 0 no correlation
- r = -1 perfect negative linear correlation
- r = 1 perfect positive linear correlation
- Linear relationship

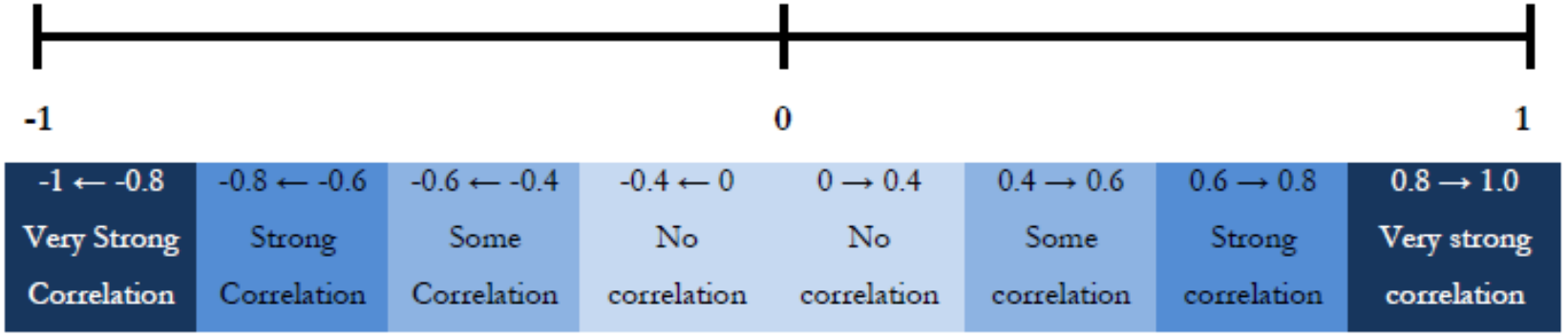
  r > 0.7: Strong positive

  0.3 < r < 0.7: Moderate

  r < 0.3: Weak

Relationship between (numerical - numerical) variables

- All correlation coefficients between 0 and 1 represent positive correlations, while all coefficients between 0 and -1 are negative correlations



Relationship between (numerical - numerical) variables

Find out the relationship between two variables (x and y).

- If highly associated
  - both two variables are predictors =>

    either variable x or y can be used in further analysis

  - One predictor and one target variable =>

    the target variable is dominated by this predictor, no need to construct complex models.

Relationship between (numerical - numerical) variables

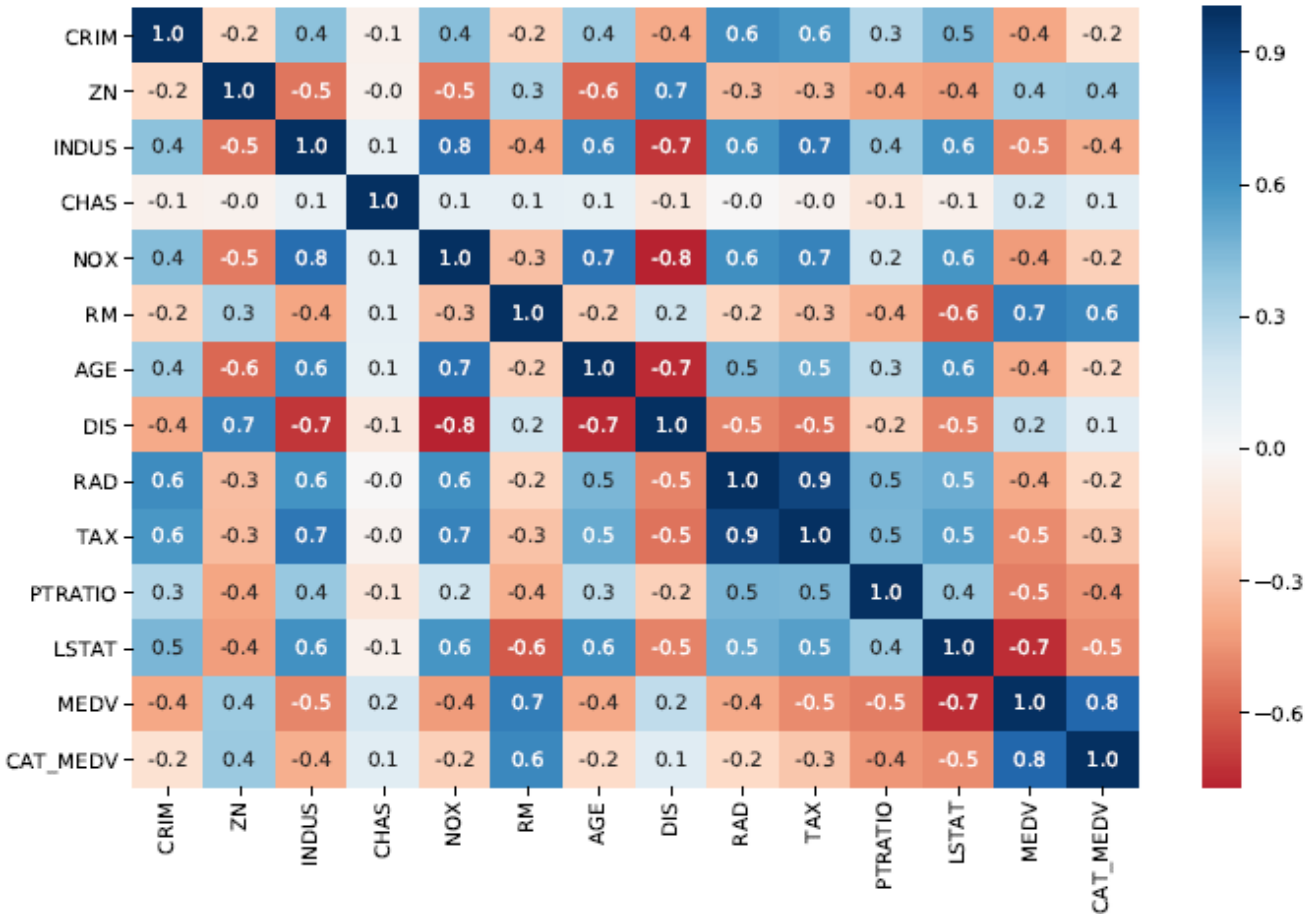
Attributes	Model	Price	Age_08_04	Mfg_Month	Mfg_Year	KM	Fuel_Type
Model	1	-0.647	0.732	-0.010	-0.735	0.313	0.103
Price	-0.647	1	-0.877	-0.018	0.885	-0.570	-0.053
Age_08_04	0.732	-0.877	1	-0.123	-0.984	0.506	0.098
Mfg_Month	-0.010	-0.018	-0.123	1	-0.057	-0.021	-0.052
Mfg_Year	-0.735	0.885	-0.984	-0.057	1	-0.505	-0.089
KM	0.313	-0.570	0.506	-0.021	-0.505	1	-0.417
Fuel_Type	0.103	-0.053	0.098	-0.052	-0.089	-0.417	1

- Price and age are highly negatively correlated
  - it is reasonable that older cars are cheaper
- Price and Kilometers also appear negatively correlated
  - cars with higher mileage are cheaper

Heatmap to highlight correlations

Darker and bluer = stronger positive correlation

Darker & redder = stronger negative correlation



Bivariate Analysis Categorical vs Categorical Analysis

- **Visualization:**
  - **Stacked Bar Chart**: categorical distribution
  - **Grouped Bar Chart**: compare between category
  - **Heatmap**
- **Contingency Table (Cross-tabulation)**
  - Matrix that displays the frequency distribution of variables.
  - It shows the counts of observations that fall

  into each combination of categories of the two variables.

	Target_0	Target_1	Total
Group_A	150	50	200
Group_B	75	125	200
Total	225	175	400

- **Chi-Square Test of Independence:**
  - Test of Independence assesses

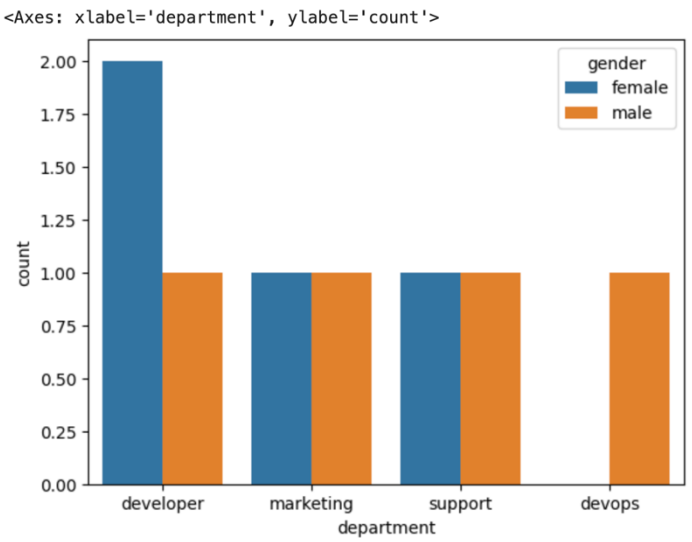
  whether there is a significant association between two categorical variables.
- **Relative Frequencies**
  - Relative frequencies show the proportion of observations in each category combination relative to the total number of observations.
  - This helps to identify trends and patterns in the data.

Categorical distribution:

Its distribution with other categorical variables

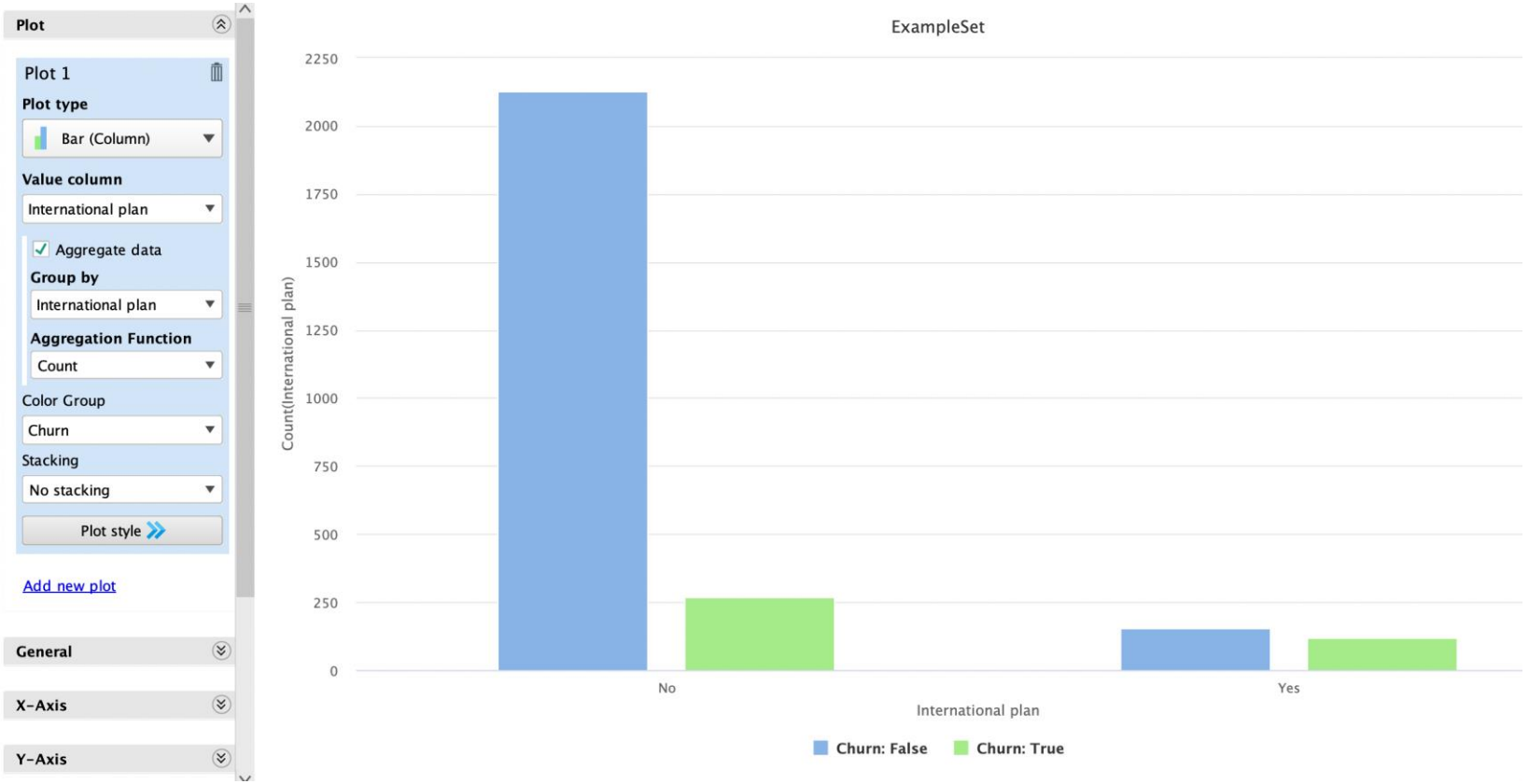
	eid	name	gender	department	age	salary	working_years	birth_place
0	E011	Anda	female	developer	39	64200.0	4	Bangkok
1	E012	Bordin	male	developer	25	48700.0	2	Phuket
2	E013	Chantana	female	developer	29	45500.0	3	Chonburi
3	E014	Donlaya	female	marketing	39	72600.0	8	Chonburi
4	E015	Ekkasit	male	marketing	37	80500.0	7	Suphanburi

```
[6] sns.countplot(data=df, x="department", hue="gender")
```



- Each department has approximately an equal number of males and females.
- However, in the **developer** department, there are more females than males,
- while the **devops** department consists only of males.

Categorical variables: Study their relation with Target



Clearly, those who have selected the **International Plan have a greater chance of leaving** the company's service than do those who do not have the International Plan.

Bivariate Analysis: Numerical vs Categorical Analysis

Analyzing to see whether the numerical variable changes or has different patterns in each group of the categorical variable.

Group Comparison Methods:

- **Central Tendency:**
  - **Group Means**:
  - **Group Medians**
- **Statistical Tests:**
  - **t-test**: compare 2 groups
  - **ANOVA**: compare 3+ groups

Visualization:

- **Grouped Boxplot**: distribution of each group
- **Violin Plot**: Boxplot + Density curve
- **Group Histograms**: histogram by category

Business Example:

”Employees in the Marketing department earn an average salary of 45,000 baht, which is higher than the IT department’s average of 38,000 baht”

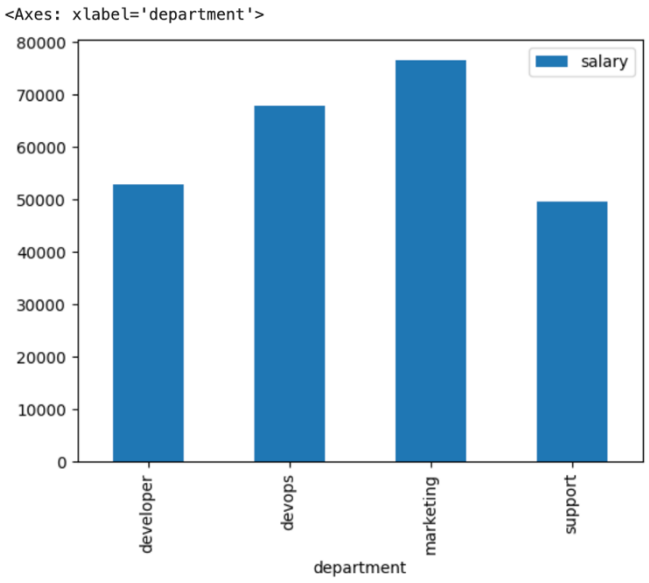
Bivariate Analysis: Numerical vs Categorical Analysis

- **Group Mean**
  - Calculating the mean of a numerical variable in each group of a categorical variable to see how the means differ between different groups.
- **Grouped Box Plot**
  - shows the distribution of a numerical variable in each group.
  - It displays the median, minimum, maximum, and the interquartile range of the data.
- **Grouped Scatter Plot**
  - show the relationship between two numerical variables.
  - It can use different colors or shapes to differentiate between groups of a categorical variable.
- **Median and Percentiles**
  - Calculating the median and various percentiles (such as 25th and 75th percentiles) to see the distribution of data in each group.
- **Frequency Distribution**
  - Histogram Showing the distribution of a numerical variable in each group by creating histograms separated by group to see the data distribution patterns.
- **Density Plot**
  - Estimating the density distribution of a numerical variable in each group by using a Density Plot to compare the distribution patterns between groups.
- Etc.

Relationships between numerical and categorical variables

	eid	name	gender	department	age	salary	working_years	birth_place
0	E011	Anda	female	developer	39	64200.0	4	Bangkok
1	E012	Bordin	male	developer	25	48700.0	2	Phuket
2	E013	Chantana	female	developer	29	45500.0	3	Chonburi
3	E014	Donlaya	female	marketing	39	72600.0	8	Chonburi
4	E015	Ekkasit	male	marketing	37	80500.0	7	Suphanburi

```
[7] dfg = df.groupby("department")[["salary"]].mean().reset_index()
    dfg.plot.bar(x="department", y="salary")
```

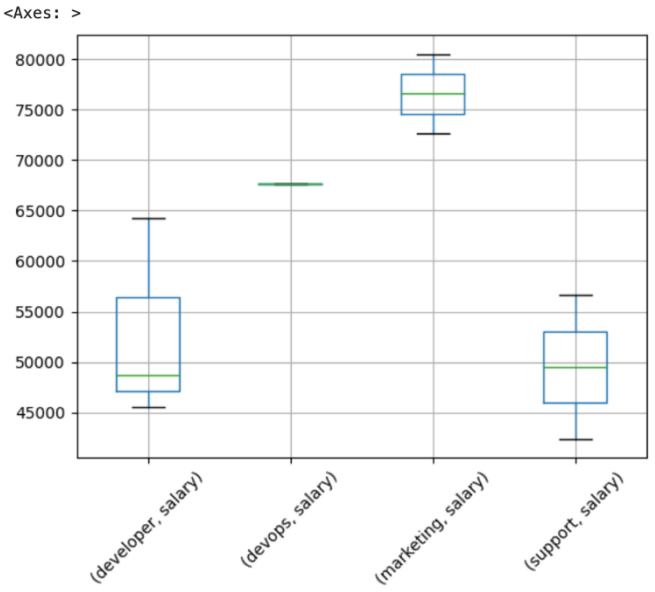


- The data is grouped by the **department**, and the average salary for each group is calculated.
- The result is displayed as a bar chart with the x-axis representing the **departments** and the y-axis representing the **average salary**.
- The **marketing department** has the highest average salary.

Relationships between numerical and categorical variables

	eid	name	gender	department	age	salary	working_years	birth_place
0	E011	Anda	female	developer	39	64200.0	4	Bangkok
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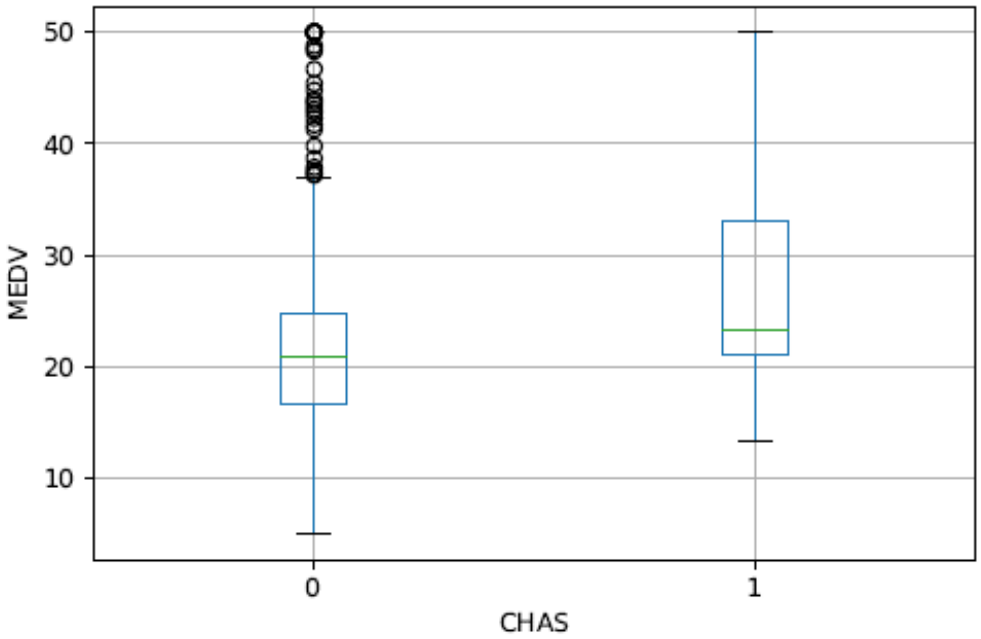
```
[8] dfg = df[["department", "salary"]].groupby("department")
    dfg.boxplot(rot=45, subplots=False)
```



- A grouped box plot shows the distribution of a numerical variable within each group.
- The **marketing** department has a higher **salary** level with less variation in its distribution.

Relationships between numerical and categorical variables

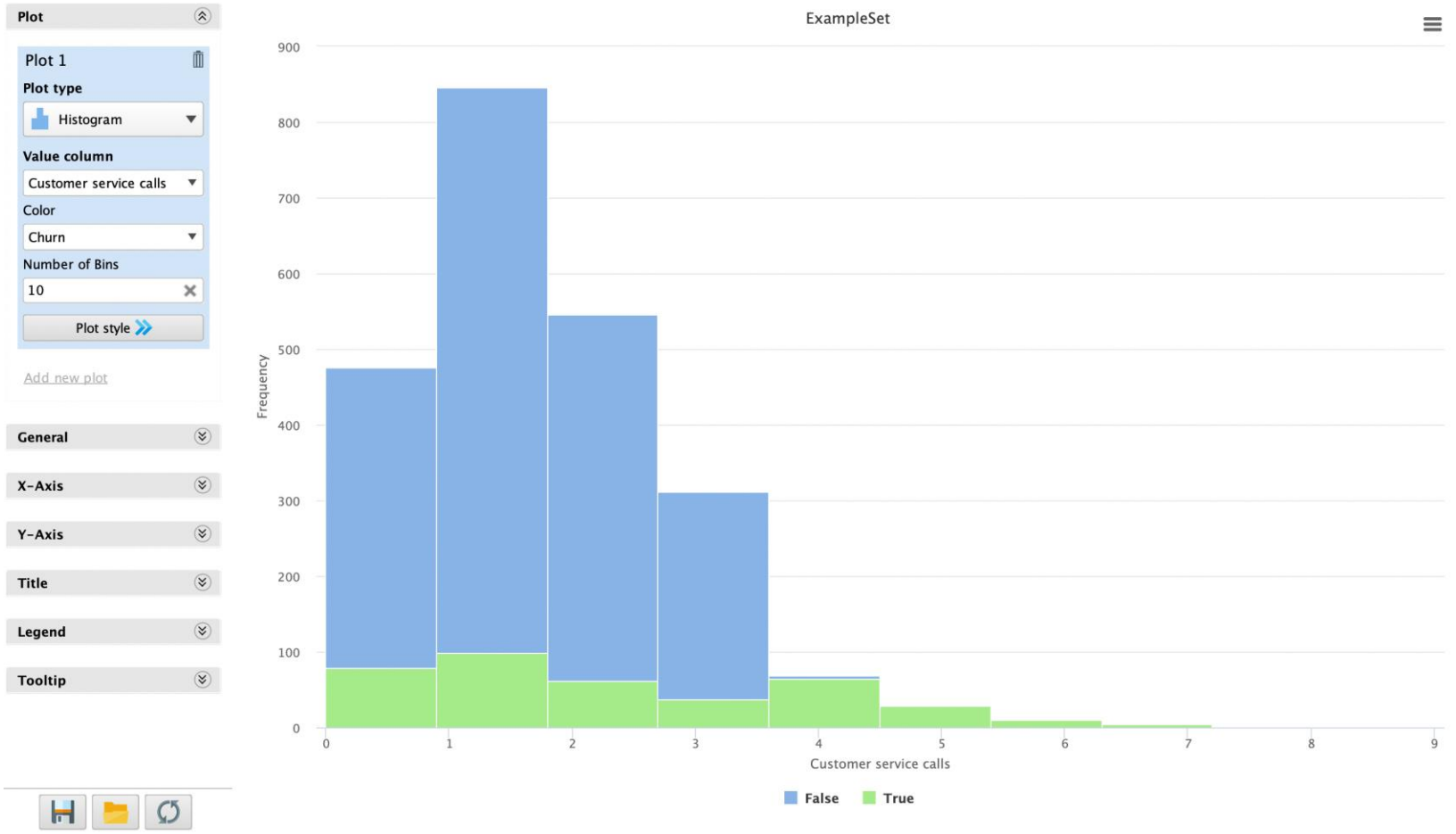
Side-by-side boxplots are useful for comparing subgroups



Houses in neighborhoods on Charles river (1) are more valuable than those not (0)

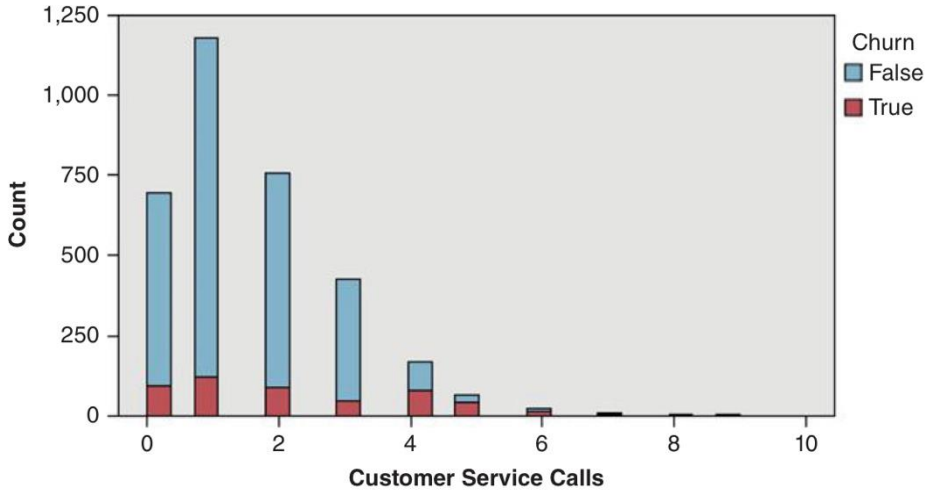
```
ax = housing_df.boxplot(column='MEDV', by='CHAS')
ax.set_ylabel('MEDV')
plt.suptitle('')  # Suppress the titles
plt.title('')
```

Histogram of customer service calls, with churn overlay

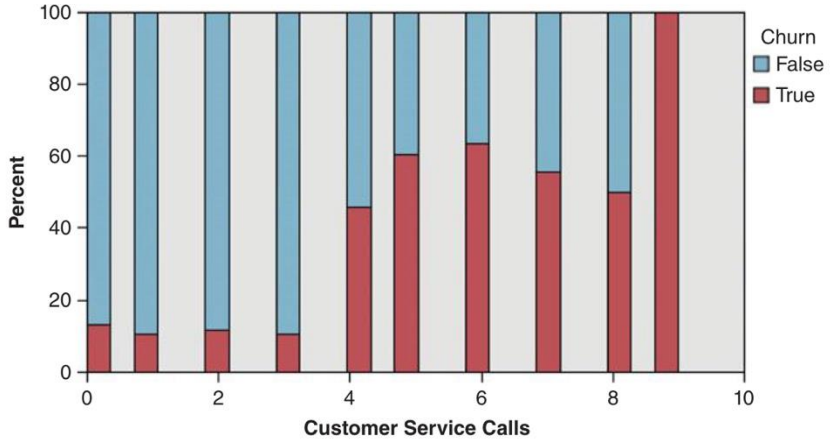


Customers who have called customer service <u>three times or less have a markedly lower churn rate</u> than customers who have called customer service four or more times

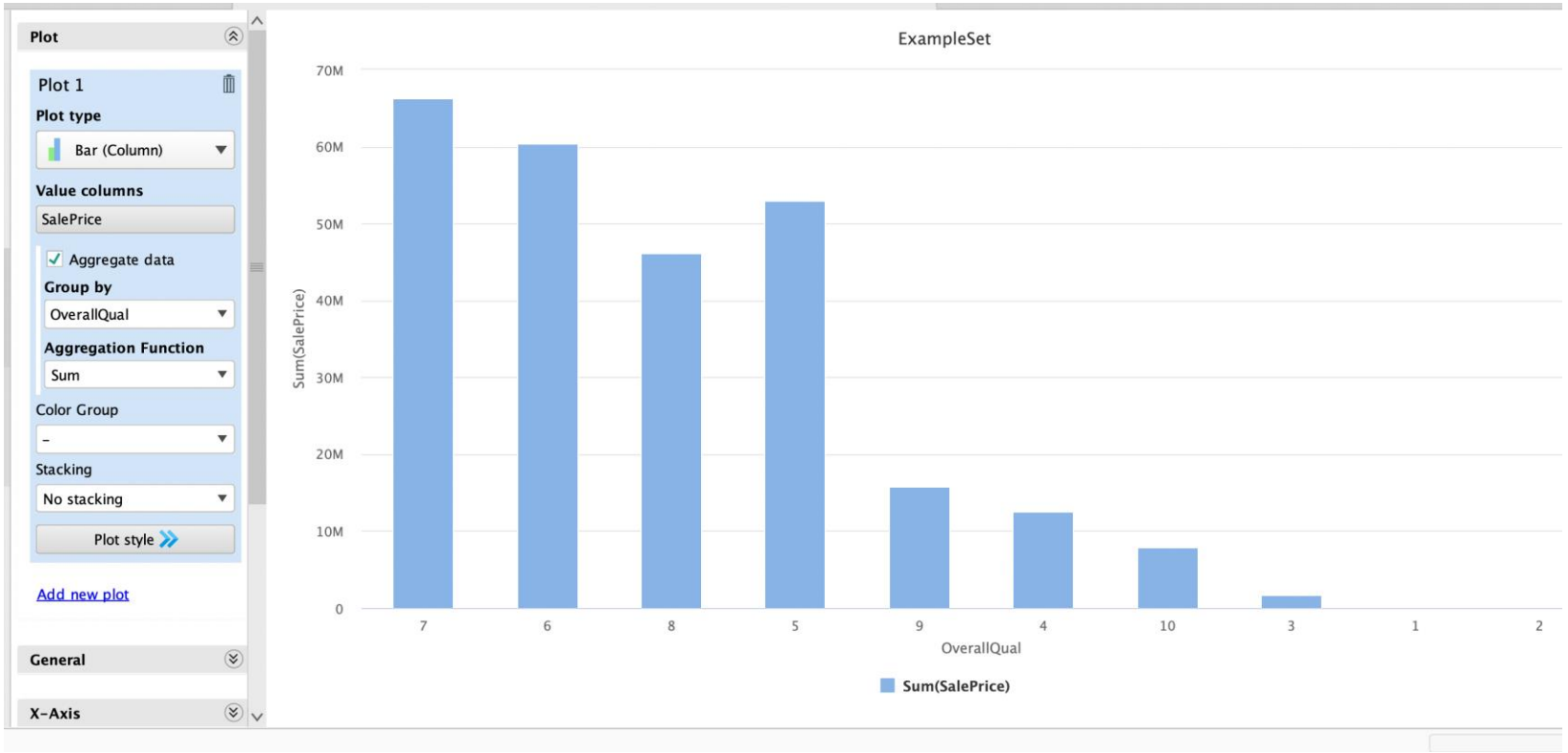
Histogram (Normalized) of customer service calls, with churn overlay



Customers who have called customer service three times or less have a **markedly lower churn rate** than customers who have called customer service four or more times



Categorical variables with Numerical Target



There is relationship between Overall-Qual with Sale-Price

Further analysis => Modelling, ...

Multivariate Analysis

Visualization Techniques:

- **Scatter Plot with Color**: 2 numerical + 1 categorical
- **Bubble Chart**: 3 numerical variables (x, y, size)
- **3D Scatter Plot**: 3 numerical variables

Advanced Techniques:

- **Correlation Matrix**: Pairwise relationships between variables
- **Principal Component Analysis (PCA)**: reduce dimension
- **Feature Interaction**: A × B, A + B combinations

Target Variable Analysis:

- **Feature Importance**: Which variable has the greatest impact on the target?
- **Segmentation**: grouping based on multiple features
- **Pattern Recognition**: find complex patterns

EXPLORING MULTIVARIATE RELATIONSHIPS: Scatterplot with color added

For classification,

2 numerical predictors &

1 categorical outcome

NOX vs. LSTAT

Red = low median value

Blue = high median value

